

SUNEESH SIDHARTHAN

ENTERPRISE AI SYSTEMS ARCHITECT

AI Governance · RAG Systems · Agentic Pipelines · Full-Stack AI Product Engineering

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Architecting enterprise-grade AI systems with strong governance, retrieval grounding, permission-aware security, and production reliability across the complete AI product lifecycle.

PROFESSIONAL SUMMARY

Enterprise AI Systems Architect and Product Engineering Leader with 15+ years designing and delivering scalable mobile, web, analytics, and cloud-native platforms across enterprise and startup ecosystems.

Specialized in production-grade AI systems: RAG architectures, multi-LLM orchestration, agentic workflows, AI-enabled mobile applications, enterprise knowledge systems, and secure AI automation platforms.

Founding architect and primary engineering lead behind MiVA — a multi-year AI-native education platform built on Kotlin/Android, Supabase/PostgreSQL, FastAPI, multi-model AI orchestration, and scalable mobile-first delivery infrastructure. Author of Sealed Enterprise Architecture (SEA 1.0) — a published 58-page foundational whitepaper defining a first-principles architectural philosophy for systems that must outlive their technologies, teams, and owners. Creator of SEAL v3, a validated implementation of SEA principles.

Proven across the complete AI product lifecycle: architecture → development → deployment → optimization → governance.

WHAT I DELIVER FOR CLIENTS

- ▶ Enterprise RAG Systems — Permission-aware internal knowledge assistants with audit logging, retrieval grounding, and model-agnostic adapter interfaces. Stack: FastAPI + pgvector + Supabase RLS + Claude/GPT.
- ▶ Agentic Workflow Automation — AI agents that read, decide, create tickets, update CRM, and trigger notifications. Built on LangGraph and tool-calling patterns.
- ▶ AI Mobile Applications — Production Kotlin/Jetpack Compose apps with LLM integration, offline support, and enterprise authentication.
- ▶ Enterprise AI Integrations — Connecting AI systems to Slack, Notion, Confluence, Jira, Salesforce, and internal APIs with full RBAC and audit trails.
- ▶ AI Governance & Drift Control — SEA-based audit logging, PII protection, model-swap-ready adapters, and architectural longevity design. Apply the 14 SEA failure mode catalogue to diagnose and remediate existing AI systems.
- ▶ Multi-LLM Content Pipelines — Automated generation, review, and production workflows across Claude, Gemini, and voice synthesis tooling.

CORE SPECIALIZATIONS

Retrieval-Augmented Generation (RAG) · Agentic AI Systems · Multi-LLM Orchestration · Enterprise Knowledge Assistants · AI Workflow Automation · AI Governance & Safety · Sealed Enterprise Architecture (SEA) · AI Drift Governance · Enterprise Architecture Longevity · Semantic Search Systems · AI Product Architecture · Prompt Pipeline Engineering · AI-Enabled Mobile Applications · Enterprise AI Integrations · Secure AI Infrastructure · Full-Stack AI Product Delivery

TECHNICAL STACK

AI / GenAI: OpenAI GPT-4.x, Claude (Anthropic), Gemini, RAG Architectures, Agentic Pipelines, Prompt Engineering, Semantic Retrieval, HITL Systems, AI Governance

Languages: Python, Kotlin, TypeScript, JavaScript, C#, SQL, HTML5, CSS

Frameworks: FastAPI, React, Node.js, Android (Kotlin/Jetpack Compose), ASP.NET MVC, Supabase

Databases: PostgreSQL, Supabase (multi-schema, RLS, pgvector), MS SQL Server, Vector DB integrations

Cloud & DevOps: Docker, GitHub Actions, CI/CD, AWS (Foundational), Railway

Methodologies: Agile/Scrum, Product-Led Engineering, AI-Native SDLC, SEAL v3 Governance

PROFESSIONAL EXPERIENCE

AI Product Architect & Engineering Lead — **MiVA — AI-Native Adaptive Learning Platform** 2023 – Present · Bengaluru

Led architecture and core platform engineering for MiVA across backend, mobile, AI orchestration, security, and deployment — from concept to Play Store closed beta.

Enterprise AI Architecture

- ▶ Designed scalable multi-schema PostgreSQL backend spanning api, app, billing, content, control, people, and social domains — structured for enterprise-grade data isolation and role-scoped access from the outset.
- ▶ Architected end-to-end platform infrastructure across mobile, backend, AI orchestration, content delivery, and security domains with extensible paths for multi-agent workflow expansion.

SEAL v3 — Validated Implementation of Sealed Enterprise Architecture (SEA 1.0)

- ▶ Designed and implemented SEAL v3: a five-ring architectural governance framework — a production-validated implementation of SEA 1.0 principles — protecting against architectural drift across all release cycles.
- ▶ Self-describing headers — every sealed file carries a 🔒 SEALED header with freeze date and human-readable enforcement warning.
- ▶ Physical sealed/ directory — bright-line boundary between frozen contracts and mutable extension layer.
- ▶ Git pre-commit hook — blocks any commit touching the sealed/ directory; overrides require documented audit trail.
- ▶ Build hash verification — SHA256 hashes of all sealed files checked at build time; any modification fails the build.
- ▶ Runtime integrity — SealedContractVerifier runs at app startup, validating function count and names against embedded manifest.
- ▶ 19 sealed API functions in SealedApi.kt — eternal contracts, never modified, never removed. New capability added via named extensions in ExtendedApi (mutable/ directory) only.
- ▶ Authority hierarchy: DATABASE > DECISION LOG > SSOT > Code > Console > Docs — single source of truth enforced at every layer.
- ▶ Published AI-Native SDLC Framework: 7-phase governance methodology (FRAME → SEAL → SPEC → BUILD → VERIFY → RECONCILE → EVOLVE) with Master Build Guide, Case Study, and Drift Audit Methodology. Released on GitHub under CC BY 4.0.

Multi-LLM & Agentic Systems

- ▶ Designed model-agnostic orchestration pipelines integrating Claude, Gemini, and future LLM providers through adapter-based abstractions — no vendor lock-in by design.
- ▶ Built agentic prompt pipelines for session generation, automated review, content refinement, production splitting, and AI-assisted sequencing — reducing production lifecycle by approximately 60%.

Enterprise Security

- ▶ Designed authentication architecture with role-based access controls, row-level security (RLS), single-device enforcement, and OTP-based auth — maintaining strong security posture across 40+ tables and 85+ production policies.
- ▶ Implemented permission-aware data architecture enforcing no cross-role data exposure at the database layer; all retrieval paths filtered before application logic.

Mobile AI Platform

- ▶ Built ExoPlayer-based adaptive video delivery with 300MB intelligent cache, custom SRT subtitle overlay (rich text, disabled native SubtitleView), and session-level state machines for seamless reel navigation.
- ▶ Engineered production resilience: CancellationException guards (44 guards / 8 files), launchSafely coroutine patterns (17 call sites), and systematic vulnerability hardening across 84 Kotlin source files.
- ▶ Integrated Razorpay payments, BunnyCDN/Cloudflare video delivery, ElevenLabs voice synthesis, and automated subtitle generation workflows.

Governed, hardened, multi-LLM — built as founding engineer to Play Store production standards across 84 Kotlin source files and multiple closed beta releases.

Solution Delivery & Digital Transformation Lead — Tata Consultancy Services (TCS) 2021 – 2023 · Bengaluru

- ▶ Managed end-to-end delivery of Global Digital Analytics & CX transformation programs for enterprise clients — from requirements through production deployment.
- ▶ Built KPI measurement frameworks and predictive analytics models for CX optimisation at enterprise scale.
- ▶ Led cross-functional teams across engineering, analytics, and business stakeholders under enterprise governance and delivery controls.

Senior Analyst — Digital Banking & Customer Analytics — Cognizant 2014 – 2020 · Bengaluru

- ▶ Led product strategy for digital banking platforms serving 10M+ users.
- ▶ Designed A/B testing frameworks delivering 25% conversion uplift; built predictive models for customer behaviour and risk assessment in regulated financial environments.

Senior Analyst — Enterprise Analytics — Dell Technologies 2012 – 2014 · Bengaluru

- ▶ Implemented enterprise analytics instrumentation; designed segmentation strategies and BI dashboards supporting executive decision-making at scale.

Software Engineer — **Mphasis** · **National Informatics Centre (Govt. of India)** 2006 – 2012 · Bengaluru / Trivandrum

- ▶ Built enterprise reporting and analytics applications for marketing and digital operations environments.
- ▶ Developed examination management systems and official web portals for DHSE, Government of India (NIC).

PRODUCT PORTFOLIO

MiVA Android App — Closed Beta — Play Store

2023 – Present ·

Production Android application delivering CBSE Class 10 Science through Instagram-style vertical video reels, powered by a multi-LLM content pipeline.

- ▶ Kotlin / Jetpack Compose · Supabase/PostgreSQL (7 schemas, 85+ RLS policies) · ExoPlayer · Razorpay · Hilt DI
- ▶ 84 Kotlin source files · SEAL v3 five-ring governance · 19 sealed API contracts · SealedContractTest CI
- ▶ Multi-LLM pipeline: Claude + Gemini + ElevenLabs voice synthesis · automated subtitle generation
- ▶ OTP auth · single-device enforcement · 300MB intelligent video cache · 40MB per-part cap · BunnyCDN delivery
- ▶ Min SDK 24 (Android 7.0) · Target SDK 35 · Kotlin 2.0.21 · Jetpack Compose BOM 2024.11

MiVA Education Web Platform — mivaedu.com

2024 – Present ·

Fully static, offline-capable learning platform for Indian students across CBSE and Kerala SCERT curricula. No login, no app, no cost.

- ▶ CBSE Class 8 AI Module — 4 chapters, 5 interactive labs, India AI Map (8 state pins), 21 flashcards, 27 glossary terms, 8 quiz questions, completion certificate. 741KB total. Works fully offline.
- ▶ CBSE Class 8 CT Module — 7 chapters, Decomposition Explorer lab, Binary Visualiser lab (0–255), fill-in-blank exercises, MCQs, exam prep with model answers, mini projects.
- ▶ Kerala SCERT ICT Standard X — Chapter 1: Design Factory / Inkscape. Horizontal snap-scroll lab pattern, bilingual EN/Malayalam toggle, ARIA accessibility throughout.
- ▶ Architecture: vanilla HTML/CSS/JS only · XOR-encoded contact · Web3Forms booking · cross-module navigation · teal #0B6E4F / navy #005F7F design systems

Sealed Enterprise Architecture (SEA 1.0) — Foundational Whitepaper

April 2026 · CC BY 4.0

A 58-page foundational whitepaper defining a first-principles architectural philosophy for systems that must outlive their technologies, teams, and owners. Written for enterprise architects, CIOs, and CTOs responsible for systems with multi-decade lifetimes.

- ▶ 7 Principles · 5 Architectural Layers (Contract, Intent, Verification, Governance, Evolution) · 14 named Failure Modes · 32 gaps in current practice · Enterprise adoption pathway mapped to TOGAF, ISO 27001, NIST AI RMF, Clean Architecture, and Domain-Driven Design.
- ▶ MiVA Education platform cited as Annex A production validation case. Part of a 5-document methodology set alongside the AI-Native SDLC, Master Build Guide, Drift Audit Methodology, and MiVA Case Study. Released on GitHub under CC BY 4.0.

AI-Native SDLC Framework — github.com/suneesh-sidharth · CC BY 4.0

A 7-phase AI development lifecycle governance methodology covering: FRAME → SEAL → SPEC → BUILD → VERIFY → RECONCILE → EVOLVE.

- ▶ Companion documents: Master Build Guide · MiVA Case Study · Drift Audit Methodology
- ▶ Positioning: governance overlay complementing GitHub Spec Kit and AWS Kiro — not competing. Applicable to any LLM-assisted solo or team development workflow.

EDUCATION & CERTIFICATIONS

Master of Computer Applications (MCA)

University of Calicut, Kerala · 2005

B.Sc Mathematics & Statistics

University of Calicut, Kerala · 2001

Executive Programme in Strategic Digital Marketing

University of Cambridge · 2019

AVAILABLE FOR

Remote freelance contracts (global) · Enterprise AI consulting · Long-term retained engagements · UK / EU market · AI governance advisory